Module 7: Communication — How Brains Connect (and Why Texting Is So Confusing)

Learning Objectives

1. Understand that communication is the process of **aligning mental models** between people — trying to get your brain's picture into someone else's brain.
2. Learn why **miscommunication** happens so often, especially through text, and how active inference explains it.
3. Develop strategies for **clearer communication** by reducing the prediction errors between what you mean and what others understand.

Introduction

You send your friend a text: "Ok." Just two letters. But depending on who reads it, that "Ok" could mean completely different things. It could be a casual agreement. It could be passive-aggressive. It could mean you're upset. It could mean you're busy. Your friend reads "Ok" and their brain has to *predict* what you meant — and sometimes they get it totally wrong.

Welcome to the weirdest prediction problem you face every day: communication. When you talk to someone, you're not directly transferring information from your brain to theirs. You're taking your internal thoughts, converting them into words (or texts, or facial expressions, or memes), sending those signals out, and hoping the other person's brain reconstructs something close to what you originally meant.

It's like a game of telephone, except it's happening in every conversation you have. Understanding how this works — and how it breaks down — might be the most useful thing you learn in this entire course.

Key Concepts

1. Communication = Model Sharing

When you communicate, you're trying to **share your mental model** with someone else. The problem is, you can't just copy and paste your thoughts into their brain. You have to encode your ideas into signals — words, tone, facial expressions, emojis — and then the other person has to decode those signals using *their* mental models.

Here's where it gets tricky. The other person's mental models are different from yours. They have different experiences, different contexts, different assumptions. So the same signal can get decoded completely differently.

Example: You tell your parent, "My day was fine." In your model, "fine" means "nothing special happened, it was normal." In their model, "fine" might mean "something is wrong but they don't want to talk about it." Same word, different models, different interpretation. Now you're in a conversation about "what's wrong" when nothing was wrong in the first place.

2. Why Text Makes Everything Harder

Face-to-face communication gives your brain tons of data to work with: words, tone of voice, facial expressions, body language, timing. Your brain uses ALL of this to predict what someone means.

Text strips away almost everything except the words. No tone. No facial expression. No body language. So your brain has to fill in the gaps with predictions, and those predictions are based on YOUR current mood and models, not the sender's.

If you're feeling insecure and someone texts you "lol sure," you might read it as sarcastic. If you're feeling confident, the same text reads as friendly. The words didn't change — your brain's predictions did.

This is why: - Sarcasm almost never works in text - Short responses feel rude even when they're not meant to be - People interpret emoji differently (is that slightly smiling face happy or passive-aggressive?) - Important conversations should happen in person when possible

3. Reducing the Prediction Gap

Good communicators don't just say things — they think about how their message will be *received*. They try to minimize the gap between what they mean and what the other person's brain will predict.

Strategies that actually work: - **Be specific**: Instead of "that was weird," say "I was surprised when you walked away during our conversation." - **Check understanding**: "Does that make sense?" or "What I'm trying to say is..." - **Match the medium to the message**: Important stuff deserves a real conversation, not a text. - **Consider their model**: What does this person know? What are they probably assuming? What context are they missing?

You can also reduce miscommunication by being a better **receiver**. When you get a message that makes you feel a strong emotion, pause and ask: "Am I reacting to what they actually said, or am I reacting to what my brain predicted they meant?"

Try It!

The Translation Experiment. Pick a conversation from your texts where something got miscommunicated or misunderstood. (Everyone has at least one.) Analyze it like a scientist: What did person A mean? What did person B's brain predict? Where was the gap? How could it have been said differently to reduce the prediction error? Share your analysis with a friend or write it down.

Summary

Communication is the process of **sharing mental models** by encoding thoughts into signals and hoping the other person decodes them correctly. **Miscommunication** happens when the sender's meaning and the receiver's prediction don't match — which is why texting is a minefield. You can improve communication by being **specific, checking understanding, choosing the right medium**, and pausing to separate what someone said from what your brain thinks they meant. Up next: **Planning** — how your brain uses everything it's learned to prepare for the future.